

Evidence: studio model cards

The shadow-size studio

Name _____ Date _____

Constructed ideal point-source model. Source and screen stay 100 cm apart. The same 4 cm-high card stays upright.

Card	Source → card distance	Model shadow height
A	20 cm	20 cm
B	40 cm	10 cm
C	50 cm	8 cm

These numbers are model predictions, not class measurements. Real LED results can differ because the source has size and edges may be blurred.

W1. Describe the pattern as source-to-card distance increases.

W2. Choose A, B or C for a 10 cm shadow. Explain.

Observation option: write a real distance and shadow height here. Label it “our measurement”.

Route 1 • Match and choose

Use the shared model table. Word bank: larger / smaller / source.

S1. Which card position gives the largest model shadow?

A. A: 20 cm

B. B: 40 cm

C. C: 50 cm

S2. Copy the model shadow height at 50 cm.

S3. In this setup, moving towards the source makes the shadow ____.

S4. Name two things kept the same.

S5. Which position gives the target 10 cm shadow?

A. 20 cm

B. 40 cm

C. 50 cm

Route 2 • Record and recommend

Make a studio instruction from the model evidence.

Task	Source → card	Model shadow height
T1	20 cm	
T1	40 cm	
T1	50 cm	

T2. Describe the pattern using two values.

T3. Recommend a position for a 10 cm shadow. State the controls.

T4. Explain why a real torch result might not match exactly.

Route 3 • Predict between the trials

The same model now places the card 30 cm from the source. Source and screen remain fixed.

X1. Predict whether its shadow is larger or smaller than the 40 cm-position shadow.

X2. Use the table to give two bounds for the predicted height. No formula is needed.

X3. Explain why the card itself has not grown.

X4. A team moves both the card and screen. Can it use our pattern without checking the new setup? Explain.

Exit and reflection

Point, draw, write or tell an adult.

E1. What happens to shadow size as this card moves towards the source?

E2. Name one thing kept fixed.

E3. Which model distance gives a 10 cm shadow?

R1. Which idea did you improve after hearing evidence?

R2. What would you investigate or ask next?
